SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 3 – COMPARITIVE ANALYSIS TASK

Course Code:	MLA0407	**Course Title:**	**Deep Learning**
CO Assessed:	CO3	**Assessment:**	Assessment Tool 3 – Comparitive analysis task
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.***		
Student Name:	CH. Durga Prasad	**Reg. No.:**	192425305
Section / Batch:	AIML	**Date:**	22.8.2026

Code of Conduct

I CH. Durga Prasad (192425305) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: CH. Durga Prasad

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Comparative Analysis Task Duration: 1hr

Comparative Analysis Task – CNN

1. CNN vs Traditional Neural Networks (ANN)

Feature	CNN	Traditional ANN
Architecture	Convolution, pooling and fully connected layers	Mainly fully connected layers
Feature extraction	Automatically learns image features	Usually requires flattening and may need manual feature engineering
Parameters	Fewer due to local connectivity and parameter sharing	Very large number of parameters
Spatial information	Preserves spatial relationships	Spatial information is mostly lost after flattening
Computation	Efficient for images	Computationally expensive for large images
Image suitability	Highly suitable	Less suitable

Conclusion: CNNs are preferred for image classification because they automatically learn spatial features such as edges, textures, and shapes while using fewer parameters.

2. Convolution Layer vs Fully Connected Layer

Feature	Convolution Layer	Fully Connected Layer
Connectivity	Local connectivity	Every neuron connects to all inputs
Parameters	Fewer due to shared kernels	Large number of parameters
Spatial information	Preserves spatial structure	Spatial structure is not preserved
Feature extraction	Extracts edges, textures and patterns	Combines extracted features for classification
Location awareness	Maintains local information	Does not directly maintain location
Role	Feature extraction	Final decision/classification

Why both are required: Convolution layers extract useful features from the image, while fully connected layers use these learned features to classify the image into different classes.

3. Max Pooling vs Average Pooling

Feature	Max Pooling	Average Pooling
Operation	Selects maximum value	Calculates average value
Main purpose	Keeps strongest features	Keeps overall feature information
Feature emphasis	Strong	Smooth/overall
Noise	Can ignore weak noise	Can include noise

Information loss	Can lose smaller features	Can reduce strong feature responses
Common use	Object/image recognition	Feature smoothing and global pooling

Max Pooling :- For

a 2 × 2 region:

[1 5]
[3 2]

Max pooling gives 5.

Advantages: Preserves important features, reduces dimensions and provides some translation tolerance.

Limitation: May discard useful information.

Preferred for: Face/object recognition, where strong edges or distinctive features are

important. **Average Pooling :-** The same region gives:

(1+5+3+2)/4 = 2.75

Advantages: Preserves overall information and produces smoother feature maps.

Limitation: Strong features may become weaker.

Preferred for: Global Average Pooling and situations where overall activation information is more important.

4. Different Filters/Kernels in CNN

A CNN uses multiple filters because different filters learn different visual features.

Layer	**Features learned**
Shallow layers	Edges, lines, corners
Middle layers	Textures, curves, simple shapes
Deep layers	Complex patterns, objects and object parts

Examples:

- Edge filters: Detect horizontal and vertical edges.
- Texture filters: Detect repeated patterns and surfaces.
- Curve filters: Detect curved boundaries.
- Complex filters: Detect shapes such as eyes, wheels or other object parts.

Why multiple filters?

One filter cannot detect every feature in an image. Multiple filters allow the CNN to create several feature maps and learn different characteristics simultaneously.

Conclusion: Early layers learn simple features, while deeper layers combine them to form complex and meaningful representations.

5. Parameter Sharing vs Fully Connected Processing

Conventional Fully Connected Approach :- Each

input location has separate weights.

For example, if an image contains thousands of pixels, connecting every pixel to neurons requires a huge number of parameters.

CNN Parameter Sharing :-

A single kernel is applied to different locations of the image.

For example, a 3×3 filter has only:

9 weights + 1 bias = 10 parameters

The same filter is reused across the entire image.

Feature	**Parameter Sharing**	**Conventional Processing**
Weights	Shared	Separate
Parameters	Much fewer	Very high
Computation	More efficient	Expensive
Location recognition	Same feature can be detected anywhere	Less efficient
Training	Faster/easier	More difficult

Conclusion: Parameter sharing makes CNNs efficient and allows them to recognize the same object feature at different positions in an image.

6. CNN With and Without Regularization

Consider:

- Model A: No regularization
- Model B: Dropout + Data Augmentation + L2 regularization

Feature	**Model A**	**Model B**
Training performance	Usually very high	May be slightly lower
Validation performance	May decrease	Usually better
Overfitting	High risk	Reduced
Generalization	Poorer	Better

Model complexity	More prone to memorization	Better controlled

Model A

The model may achieve very high training accuracy but perform poorly on unseen images because it memorizes training examples.

Model B

- Dropout: Randomly disables neurons during training.
- Data Augmentation: Creates variations such as rotation, flipping and cropping.
- L2 Regularization: Penalizes excessively large weights.

Conclusion

Model B is more reliable for real-world image classification because regularization reduces overfitting and improves generalization to unseen images.

7. AlexNet vs ResNet

Feature	AlexNet	ResNet
Architecture	Traditional deep CNN	Residual CNN
Depth	8 learned layers	Common versions: 18, 34, 50, 101, 152
Convolutional layers	Relatively few	Many
Training difficulty	Easier for its era	Designed to make very deep networks trainable
Parameters	About 60 million	Depends on version; ResNet-50 has about 25.6 million
Vanishing gradient	More susceptible	Greatly reduced
Skip connections	No	Yes
Complex tasks	Good historically	Generally much better

ResNet Skip Connection

Instead of learning:

Output = $F(x)$

ResNet learns:

Output = $F(x) + x$

Here, x is passed directly through a shortcut connection.

Advantages of ResNet

- Reduces vanishing-gradient problems.

- Makes deeper networks easier to train.
- Improves feature learning.
- Provides better performance on complex image tasks.

Recommendation: ResNet is generally more suitable for a modern image-classification application because it supports deeper architectures and provides strong performance with residual connections.

8. Shallow CNN vs Deep CNN

Feature	Shallow CNN	Deep CNN
Number of layers	Few	Many
Feature learning	Mainly low-level features	Low-level + high-level features
Computation	Low	High
Training	Easier	More complex
Overfitting	Lower for small models	Can be higher without regularization
Accuracy	Good for simple tasks	Better for complex tasks
Resource requirement	Low	High

Shallow CNN

Can learn features such as:

Edges → Simple shapes

It is suitable for simple datasets such as handwritten digits.

Deep CNN

Can learn:

Edges → Textures → Shapes → Object parts → Complete objects

For complex image-recognition problems, deep networks are more powerful.

Why ResNet is useful:

ResNet uses skip connections to make deep networks easier to train and reduces problems such as vanishing gradients.

9. CNN Architecture Comparison for Different Applications

The same CNN architecture should not necessarily be used for all applications.

Application	Requirements	Suitable approach
Face Recognition	High accuracy, facial details, large datasets	Deep CNN/Face-specific architectures
Medical Image Classification	Very high accuracy, fine details, limited datasets	Transfer learning/ResNet-like models
Autonomous Vehicles	High accuracy + very fast inference	Efficient CNN architectures
Handwritten Digit Recognition	Simple images, low complexity	Small/shallow CNN

Factors to consider

1. Dataset size – Large datasets can support deeper models.
2. Image complexity – Complex images require stronger feature extraction.
3. Computational resources – GPUs and memory affect architecture selection.
4. Accuracy – Safety-critical applications require high accuracy.
5. Inference speed – Real-time applications need fast models.
6. Model size – Mobile/embedded systems require lightweight networks.
7. Training time – Deeper models generally require more resources.

Conclusion: Architecture should be selected according to the application's accuracy, complexity, dataset, hardware and speed requirements.

10. Complete CNN Architecture Analysis

The CNN architecture is:

Input → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Fully Connected → Output

1. Input Image

The original image is given to the CNN, usually as pixel values with dimensions such as $224 \times 224 \times 3$ for an RGB image.

2. Convolution Layer

Filters/kernels scan the image and extract features such as edges, lines and textures. The result is a feature map.

3. ReLU

ReLU is given by:

$\text{ReLU}(x) = \max(0, x)$

It removes negative values and introduces non-linearity.

4. Pooling

Pooling reduces the size of feature maps while retaining important information. Max pooling selects the strongest feature from a region.

5. Second Convolution + ReLU

The second convolution layer learns more complex features such as shapes and object parts. ReLU again adds non-linearity.

6. Second Pooling

Further reduces the feature-map size and computational requirements.

7. Fully Connected Layer

The extracted feature maps are flattened and passed to fully connected neurons. These neurons combine the learned features for classification.

8. Output Layer

The final layer produces class probabilities.

Example:

Cat = 10%, Dog = 85%, Bird = 5%

Therefore, the prediction is Dog.

CNN vs ResNet

A basic CNN passes information sequentially through layers. ResNet uses skip connections:

Output = F(x) + x

The input is directly added to the output of a block.

Advantages of ResNet:

- Reduces vanishing-gradient problems.
- Improves gradient flow.
- Makes deeper networks easier to train.
- Gives better performance on complex image tasks.

Conclusion

A basic CNN is suitable for simple image classification, while ResNet is better for complex tasks because residual connections make deep networks easier to train and improve performance.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________ Date: ________________	Signature: ________________ Date: ________________	Signature: ________________ Date: ________________